

6(a)

any **two** of these pairs, feature and explanation

- 1 thick wall of (left) ventricle ; **R** if right
 - 2 pumps blood at high pressure to, reach (all of) systemic circulation / overcome resistance in systemic circulation / over a long distance / AW ;
A if thick wall unqualified **A** generate high force
R 'withstands high pressure'
 - 3 thin wall of (left) atrium ;
 - 4 pumps blood at low pressure, into ventricle / over a short distance ;
- either**
- 5 (named) valves ; **R** tricuspid / pulmonary
 - 6 prevent, backflow of blood / maintains one-way flow of blood (through heart) ;
ecf if right atrioventricular / tricuspid
- or**
- 5 (left) atrioventricular / mitral / bicuspid, valve ; **R** right
 - 6 prevents, backflow of blood / blood flowing from ventricle to atrium ;
A maintains one-way flow of blood (through heart)
- or**
- 5 (left) semi-lunar, valve ;
 - 6 prevents, backflow of blood / blood flowing from, aorta / artery, to ventricle ;
A maintains one-way flow of blood (through heart)
- 7 chordae tendinae / tendons / tendinous cords / AW ; **I** ligaments
 - 8 hold the valve in position, during systole / when ventricle contracts / AW ;
A prevents blowback / turning inside out
I high tensile strength unqualified
 - 9,10 AVP ;; e.g
papillary muscle ;
contracts to hold, tendons / valve, when heart contracts / during systole ; **ora**

4

Question	Answer	Marks
6(b)	<i>any four from: if AVN used penalise once and then use ECF</i> <u>sinoatrial node</u> releases, waves of excitation / waves of depolarisation / (electrical) impulses / action potential(s) ; AW - impulses / AW, stimulate, (both) atria to contract / atrial systole ; - impulses / AW, prevented from reaching ventricles by, fibrous ring / non-conducting tissue / insulating tissue / (between atria and ventricles) ; A annulus fibrosus <u>atrioventricular node</u>, delays impulse (by 0.1 s) ; - atrioventricular node, sends impulse to, Purkyne tissue (in septum) ; A bundle of His <i>allow ecf for abbreviating AVN</i> - Purkyne tissue conducts impulse to, base of ventricles / apex of heart, so they contract ;	4
6(c)(i)	<i>any two from: if transport is used in MP2 to MP8, penalise once and mark to a max of 1 for those MPs</i> <i>provides, aqueous / watery, (external) environment / surroundings, for cells</i> - to maintain correct water potential to prevent, cell dehydration / excessive water loss from cells ; <i>idea that provides medium for exchange of substances</i> - supply of oxygen / oxygen diffuses from blood into tissue fluid ; - removal of carbon dioxide / carbon dioxide diffuses from tissue fluid into blood ; A exchange of (respiratory) gases as alternative to MP2 and MP3 - supply of named nutrient e.g. glucose, amino acids, (named) ion(s) ; - removal of, metabolic / toxic, waste or urea ; - location / movement, of (tissue) macrophages / neutrophils / phagocytes (for defence) ; I white blood cell - allows movement of / medium for passage of, (named) cell signalling molecules / hormones / ligands ; A ref. to cells signalling nearby cells / paracrine signalling - AVP ; e.g. contributes to formation of lymph / contains antibodies e.g. allows movement of lymphocytes into lymph	2

Question	Answer	Marks
6(c)(ii)	any three from: 1 (at arteriole end of capillary) high, hydrostatic / blood, pressure ; 2 (ultra)filtration / described ; e.g. fluid / plasma / water <u>and</u> solutes, forced / comes, out of, capillaries / blood 3 glucose / amino acids / ions, leave blood R if stated, red blood cells / platelets, leave or (large) plasma proteins remain in blood ; 4 through, fenestrations / fenestrae / endothelial pores or through, pores / gaps / spaces, in, capillary wall / endothelium / between cells ; 5 AVP ; e.g. <i>ref. to</i> hydrostatic pressure gradient hydrostatic pressure of blood is greater than hydrostatic pressure of tissue fluid	3
6(d)(i)	I pathogen A epitope for antigen any two from: some (T-lymphocyte) have (T-cell) receptors, specific / complementary, to a particular antigen ; ora antigen (in vaccine) binds to these T-lymphocytes ; <i>idea that</i> only these T-lymphocytes are, activated / stimulated / selected ; I respond AVP ; e.g. some T-lymphocytes do not become exposed to limited quantity of antigen in vaccine	2

Question	Answer	Marks
6(d)(ii)	<i>max four if no correct ref. within response to, memory cells / immunological memory</i> <i>any five from:</i> <i>context of primary immune response</i> 1 antigen presentation by macrophages ; 2 (T- / B-) lymphocyte, binding / recognition / clonal selection / activation / described in terms of receptor ; 3 clonal expansion / cell divides by mitosis many times (to produce a clone) / AW ; 4 cytokines released by T-helper, cells / lymphocytes ; 5 (stimulate) formation of memory, (T- / B-) lymphocytes / cells ; 6 (memory cells) remain in, circulation / body, for a long time / AW ; A remain in lymph nodes or memory cells are long-lived ; <i>context of secondary immune response</i> 7 large numbers of (specific), memory cells / lymphocytes ; I more memory cells A memory cells have a higher chance of encountering antigen	5